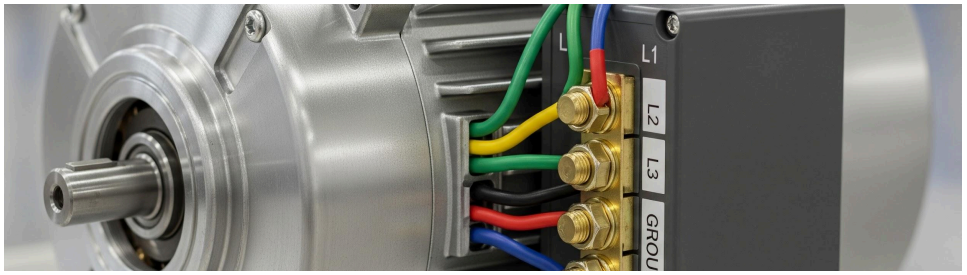


# Electrical Machine Data Sheet

### Machine Name/Type : DC Generator Coupled Set

### Name plate Details

The rating section details the machine's operational capacity and design limits.

| PARAMETER | DC Shunt Motor | DC Comp Generator | UNIT |
|---|---|---|---|
| Power output | 5 | 4 | HP |
| Voltage | 220 | 220 | V |
| Rated Current | 19 | 13.6 | A |
| Speed | 1500 | 1500 | RPM |
| Field current | 1 | 1 | A |

## Precautions

I. Ensure supply voltage is **220 V DC** before switching ON.

II. Never run the DC motor or generator **without field excitation**.

III. Always start the motor using a **starter** to limit starting current.

IV. Check proper mechanical coupling and alignment before operation.

V. Ensure both machines are properly earthed.

VI. Do not exceed rated armature current for either machine.

VII. Apply electrical load on generator **gradually**.

VIII. Avoid sudden changes in load to prevent mechanical stress.

IX. Monitor voltage, current, speed, and temperature continuously.

X. Switch OFF immediately if abnormal noise, sparking, or vibration occurs.

## Working Principle of DC Motor – Generator Set

The DC motor operates on the principle that a **current-carrying conductor placed in a magnetic field experiences a force**, producing torque and mechanical rotation.

The shaft of the motor is mechanically coupled to the DC generator. As the motor rotates, it drives the generator shaft.

The DC generator works on the principle of **electromagnetic induction**, where mechanical energy is converted into electrical energy.
Thus, electrical energy supplied to the motor is converted into mechanical energy and then back into electrical energy at the generator terminals.

## Working Characteristics

I. Smooth and steady speed under normal operating conditions.

II. Generator output voltage increases with speed and field excitation.

III. Overall efficiency depends on losses in both motor and generator.

IV. Suitable for studying load characteristics and efficiency.

V. Demonstrates clear energy conversion stages.

### Applications

1. Laboratory testing and experiments
2. Load testing of DC generators
3. Efficiency determination of DC machines
4. Educational demonstration of energy conversion

5. Variable DC power supply systems
6. Motor loading applications
7. Research and training setups
8. Speed control studies
9. Electrical machine calibration
10. Teaching electromechanical principles